Quadrilaterals: Which Way Does the Implication Go? Answer Key

High School Geometry — Polygons

Quick Answers

1. 15	3. —	5. 8
2. 2.83, 8.49	4. 3.61, 6.71	6. 4.47, 6.32, 6.32

Common wrong answers

2. *If they got 8.49: assumed a rhombus has congruent diagonals, so reported AC's length for BD*
4. *If they got 6.71: assumed perpendicular diagonals force all four sides congruent, so reported AB's length for BC*
5. *If they got 3: subtracted the 5 instead of adding it when moving it across the equals sign*

Part (a) of each problem is a computation and is machine-verified. Part (b) is a diagnosis in words: a model answer is given, together with what to accept and what not to. The sentence to insist on names *both* the property that was assumed and the property that was actually established.

1. $MNPQ$ is a rhombus with consecutive angles $(4x+10)^\circ$ and $(6x+20)^\circ$.

(a) Solution: A rhombus is a parallelogram, so consecutive angles are supplementary.

$$\begin{aligned}(4x+10)+(6x+20)&=180\\ 10x+30&=180\\ x&=15\end{aligned}$$

$$\boxed{x=15}$$

The angles are $4(15)+10=70^\circ$ and $6(15)+20=110^\circ$.

(b) Manual review — expected reasoning. No. A square is a rhombus with right angles, so the implication square $\Rightarrow$ rhombus runs one way only. This rhombus has a 70° angle, so it is not a square. *Accept* any answer that points at the angles (or at the diagonals not being congruent). *Do not accept* "no, because it does not look like one" or an answer that treats "all sides congruent" as decisive.

2. $A(1,1)$, $B(5,3)$, $C(7,7)$, $D(3,5)$, all four sides congruent. Priya: "a square, so both diagonals are about 8.49".

(a) Solution: Use the distance formula on each diagonal.

$$AC=\sqrt{(7-1)^2+(7-1)^2}=\sqrt{72}\qquad BD=\sqrt{(3-5)^2+(5-3)^2}=\sqrt{8}$$

$$\boxed{AC\approx 8.49}$$
$$\boxed{BD\approx 2.83}$$

(b) Manual review — expected reasoning. Priya's first step is sound: four congruent sides do force a rhombus. The step that fails is the next one — rhombus $\Rightarrow$ square, which is the converse of a one-way implication. The diagonals settle it: $8.49 \neq 2.83$, so $ABCD$ is not a rectangle and therefore not a square. Most specific correct name: **rhombus**. *Do not accept* "Priya is right, she just miscalculated" — the arithmetic was never the problem.

3. Always true or not always true, with a counterexample where it fails.
Manual review — expected answers.
(a) Always true. A rectangle has two pairs of parallel sides by definition, which is the definition of a parallelogram. (This is the "upward" direction of the family tree.)
(b) Not always true. A parallelogram needs right angles to be a rectangle. Counterexample: the parallelogram with vertices $(0, 0)$, $(5, 0)$, $(7, 3)$, $(2, 3)$ — opposite sides are parallel and congruent, but no angle is $90°$.
(c) Not always true. Perpendicular diagonals give a rhombus only if the diagonals also *bisect* each other. Counterexample: the kite with vertices $(0, 5)$, $(2, 0)$, $(0, -3)$, $(-2, 0)$ — its diagonals lie on the axes and so are perpendicular, but the sides measure $\sqrt{29}$ and $\sqrt{13}$, so they are not all congruent.
(d) Always true. A square is by definition a quadrilateral with four congruent sides and four right angles; four congruent sides alone is the definition of a rhombus.
Accept any correct counterexample (a described drawing is fine). *Do not accept* a bare "false" with no counterexample for (b) or (c).

4. $A(0, 6)$, $B(3, 0)$, $C(0, -2)$, $D(-3, 0)$. Sam: "perpendicular diagonals, so a rhombus with every side about 6.71".
(a) Solution:

$$AB = \sqrt{(3-0)^2 + (0-6)^2} = \sqrt{45} \qquad BC = \sqrt{(0-3)^2 + (-2-0)^2} = \sqrt{13}$$

$$\boxed{AB \approx 6.71}$$
$$\boxed{BC \approx 3.61}$$

(b) Manual review — expected reasoning. Sam is right that the diagonals are perpendicular, but that alone is not sufficient: for a rhombus the diagonals must be perpendicular *and* bisect each other. Here the diagonal AC has midpoint $(0, 2)$ while BD has midpoint $(0, 0)$, so they do not bisect each other — and indeed $AB \neq BC$. $ABCD$ is a **kite**. *Accept* either the midpoint argument or the two unequal side lengths as the disproof; the sentence must say what perpendicular diagonals alone do *not* guarantee.

5. $PQRS$ is a parallelogram, $PR = 3x - 5$, $QS = x + 11$. Sam gets $x = 3$ and says "square".
(a) Solution: Setting the diagonals equal is the right move; the slip is in the algebra. Moving -5 across the equals sign *adds* 5:

$$3x - 5 = x + 11 \;\Rightarrow\; 2x = 11 + 5 = 16 \;\Rightarrow\; x = \frac{16}{2}$$

$$\boxed{x = 8}$$

Sam computed $11 - 5 = 6$ and halved it, which is where the 3 came from. Check: both diagonals measure $3(8) - 5 = 19$ and $8 + 11 = 19$.
(b) Manual review — expected reasoning. Congruent diagonals in a *parallelogram* force a **rectangle** — that theorem is genuine. But rectangle $\Rightarrow$ square is the converse of

a one-way implication and does not follow: a 19-unit diagonal says nothing about the side lengths. To reach "square" you would need one more fact, such as two congruent adjacent sides or perpendicular diagonals. *Do not accept* "Sam is right once x is fixed".

6. Challenge: $W(-1,2)$, $X(3,4)$, $Y(5,0)$, $Z(1,-2)$, all four sides congruent.

(a) Solution:

$$WX = \sqrt{4^2+2^2} = \sqrt{20} \qquad WY = \sqrt{6^2+(-2)^2} = \sqrt{40} \qquad XZ = \sqrt{(-2)^2+(-6)^2} = \sqrt{40}$$

$$\boxed{WX \approx 4.47}$$
$$\boxed{WY \approx 6.32}$$
$$\boxed{XZ \approx 6.32}$$

(b) Manual review — expected reasoning. Four congruent sides give a rhombus; congruent diagonals in a parallelogram give a rectangle; a quadrilateral that is both is a **square**. Unlike problem 2, the diagonal test passes here, and that single test — comparing WY with XZ — is exactly the one that separates "rhombus but not square" from "square". An equivalent sufficient test is showing one angle is right, e.g. the slopes of WX and XY are $\frac{1}{2}$ and -2. *Accept* either test; the answer must name the test, not just the shape.